SOFIANE MALLEM

Python Developer

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LinkedIn | GitHub | Portfolio

SUMMARY

Python developer with production experience running data and rendering pipelines on Linux. Builds backend services with FastAPI, PostgreSQL and Redis; comfortable across backend, data and infrastructure work.

EXPERIENCE

Synthavo **Nov 2025 – Apr 2026**
Python Developer *Stuttgart, Germany*

- Owned the CAD data pipeline (Python, Pandas): validated, transformed and uploaded customer datasets feeding a Blender rendering service on 5+ Linux servers.
- Migrated rendering to a newer Blender: ∼40% faster end-to-end rendering on top of prior optimizations, 99%+ uptime via monitoring and bottleneck fixes.
- Maintained the rendering-to-training pipeline and ran YOLOv5 training end to end; delivered weights to QA.
- Ran deployments with Ansible (existing IaC); served as primary contact for production incidents and customer requirements.

Synthavo **Feb 2025 – Aug 2025**
Python Developer Intern *Stuttgart, Germany*

- Built Python tooling for CAD validation and transformation, improving rendering reliability.
- Added STL mesh reduction ahead of rendering, freeing RAM for more parallel instances and making end-to-end rendering ∼40% faster.
- Cut customer onboarding provisioning from ∼5 days to 2–3 through Python automation.

PROJECTS

Hang | *Real-Time Messaging Platform, Horizontally Scaled* **2026 · GitHub**
Stack: FastAPI · WebSockets · Redis Pub/Sub · PostgreSQL · Nginx · Docker · OAuth · S3 · Prometheus/Grafana · React/TS

- Built a dating/chat platform with Google OAuth, S3 uploads, swiping, typing indicators, read receipts and cursor pagination; owned architecture and backend.
- Synced WebSocket events across stateless FastAPI replicas behind Nginx via Redis Pub/Sub; added per-user rate limits and Prometheus/Grafana monitoring.
- Load-tested 190,000 messages on 3 replicas: zero loss at 2,000 concurrent WebSockets, p99 under 111 ms; found and fixed a DB-session leak and a join-before-accept race.

LAD (Local AI Daemon) | *Self-Hosted AI Assistant with Tool-Calling Agent* **2026 · GitHub**
Stack: FastAPI · React/TS · PostgreSQL + pgvector · Ollama · MCP · SSE · Docker Compose · Nginx · GitHub Actions

- Built a fully local AI assistant on Ollama (no external API): streaming chat, file uploads, MCP tool-calling agent with server-side per-mode tool allowlists.
- Sandboxed shell execution in an offline, non-root container with dropped capabilities and resource limits.
- Implemented cross-chat memory from embedded rolling summaries (pgvector); tested with pytest (real Postgres) and Vitest/MSW, with 3 CI workflows.

Durden | *Object Detection API on Google Cloud Run* **2026 · GitHub · Live**
Stack: FastAPI · Ultralytics YOLO · Docker · GitHub Actions · GHCR · Google Cloud Run · Workload Identity Federation

- Built a YOLO inference API with per-request model selection, background model loading and readiness endpoints.
- Set up CI that runs lint and tests, publishes to GHCR and deploys to Cloud Run via workload identity federation (no stored keys).

EDUCATION

42 Heilbronn (peer-learning software engineering program) **Feb 2023 – Nov 2025**
Software Engineering Course *Heilbronn, Germany*
University of Science and Technology Houari Boumediene (USTHB) **2017 – 2020**
Bachelor of Computer Science *Algiers, Algeria*

TECHNICAL SKILLS

Programming: Python, SQL, JavaScript, TypeScript, React, C, C++, Bash
Backend: FastAPI, REST APIs, WebSockets, OAuth, Async Python, Redis Pub/Sub
Data: PostgreSQL, Redis, Pandas, Data Pipelines
Infrastructure: Docker, Linux, Nginx, CI/CD, GitHub Actions, Google Cloud Run, Ansible, Prometheus, Grafana
Practices: Git, pytest, Load Testing, Unit/Integration Testing, Agile, AI-Assisted Development (Claude Code, Cursor)

LANGUAGES

English (Fluent) | French (Native) | German (A1, learning)